

Computer Networks Study Guide

Chapter 1: Computer Networks and the Internet

1.1 What Is the Internet?

1. What are the two principal ways to describe the Internet?

1. A nuts-and-bolts description of its hardware and software components
2. A services description as an infrastructure that provides services to distributed applications

2. In Internet jargon, what are the billions of computing devices connected to the Internet called?

Hosts or end systems

3. What are the two most prominent types of packet switches in today's Internet?

Routers and link-layer switches

4. What is the name for the sequence of communication links and packet switches traversed by a packet from sender to receiver?

A route or path

5. Through what do end systems access the Internet?

Internet Service Providers (ISPs)

6. What is the collective name for the Internet's principal protocols?

TCP/IP

7. Which organization develops Internet standards, and what are its documents called?

The Internet Engineering Task Force (IETF). Its documents are called Requests for Comments (RFCs)

8. Which committee specifies standards for Ethernet and WiFi?

The IEEE 802 LAN Standards Committee

9. From a services perspective, the Internet is an infrastructure that provides services to what?

Distributed applications

10. Where do Internet applications run?

On end systems (hosts), not in the packet switches in the network core

11. What is the name of the interface that specifies how a program asks the Internet to deliver data to a destination program on another end system?

The socket interface

12. Using a human analogy for a protocol, what might a response of "Don't bother me!" to a "Hi" indicate?

An unwillingness or inability to communicate

13. What three things does a network protocol define?

The format, the order of messages exchanged, and the actions taken on the transmission/receipt of a message or other event

1.2 The Network Edge

14. What are the components located at the edge of the Internet called?

End systems or hosts

15. What is the equation given for hosts and end systems?

Host = End system

16. Into what two categories are hosts sometimes divided?

Clients and servers

17. Where do many of the servers we use today reside?

In large data centers

18. What is the access network?

The network that physically connects an end system to the first router (edge router) on a path to other end systems

19. What are the two most prevalent types of broadband residential access discussed?

Digital Subscriber Line (DSL) and cable

20. What does a DSLAM do?

A Digital Subscriber Line Access Multiplexer (DSLAM) located in the telco's central office separates data and phone signals and sends data into the Internet

21. What does HFC stand for, and what two types of cable does it use?

Hybrid Fiber Coax. It uses fiber optics and coaxial cable

22. What is the device at the cable head end that serves a similar function to a DSLAM?

The Cable Modem Termination System (CMTS)

23. Why is cable Internet access considered a shared broadcast medium?

Because every packet sent by the head end travels downstream on every link to every home

24. What does FTTH stand for?

Fiber To The Home

25. What is the dominant wired access technology in corporate, university, and home LANs?

Ethernet

26. What is the common name for IEEE 802.11 wireless LAN technology?

WiFi

27. Compared to WiFi, what is the typical range for a user from a cellular base station?

A user need only be within a few tens of kilometers (as opposed to a few tens of meters for WiFi)

28. What are the two categories of physical media?

Guided media and unguided media

29. What is the least expensive and most commonly used guided transmission medium?

Twisted-pair copper wire

30. What is a key characteristic of coaxial cable that allows it to achieve high data rates?

Its construction with two concentric conductors, special insulation, and shielding

31. What are three advantages of fiber optics as a transmission medium?

It can support tremendous bit rates, is immune to electromagnetic interference, and has very low signal attenuation

32. What is a key reason fiber optics is not yet prevalent for short-haul transport like LANs?

The high cost of optical devices like transmitters, receivers, and switches

33. What are the three broad groups of terrestrial radio channels based on distance?

Those that operate over very short distances, in local areas, and in wide areas

34. What are the two types of satellites used in communications?

Geostationary satellites and low-earth orbiting (LEO) satellites

35. What is a significant disadvantage of geostationary satellites?

They introduce a substantial signal propagation delay of about 280 milliseconds due to their high altitude

1.3 The Network Core

36. What are the two fundamental approaches to moving data through a network?

Circuit switching and packet switching

37. In a network application, what do end systems exchange?

Messages

38. What are the smaller chunks of data that a long message is broken into called?

Packets

39. What does "store-and-forward transmission" mean?

The packet switch must receive the entire packet before it can begin to transmit the first bit of the packet onto the outbound link

40. In a simple two-end-system, one-router example, if a packet is L bits long and the link rate is R bits/sec, what is the total delay to get the packet to the destination?

$2L/R$ seconds

41. For a path with N links, each of rate R , what is the end-to-end delay for one packet?

$N \times (L / R)$ seconds (assuming only store-and-forward delay)

42. What is the purpose of an output buffer (output queue) in a packet switch?

To store packets that the router is about to send into a link when that link is busy

43. What is the variable delay that packets suffer in addition to store-and-forward delays?

Queuing delays

44. What happens if an arriving packet finds the output buffer full?

Packet loss occurs—either the arriving packet or one of the already-queued packets is dropped

45. How does a router determine which outbound link to forward a packet onto?

It examines the packet's destination address and searches its forwarding table to find the appropriate outbound link

46. What are the special protocols used to automatically set forwarding tables called?

Routing protocols

47. In circuit switching, what is reserved for the duration of a communication session?

The resources needed along the path (buffers, link transmission rate)

48. What are the two fundamental approaches to multiplexing a circuit in a link?

Frequency-division multiplexing (FDM) and time-division multiplexing (TDM)

49. What is the main criticism of packet switching regarding real-time services?

Its variable and unpredictable end-to-end delays

50. What are two arguments made by proponents of packet switching?

1. It offers better sharing of transmission capacity
2. It is simpler, more efficient, and less costly to implement

51. What is the overarching goal of interconnecting access ISPs?

So that all end systems can send packets to each other

52. Why is directly connecting every access ISP to every other access ISP not a practical solution?

It is much too costly, as it would require each access ISP to have a separate communication link to every other access ISP in the world

53. In a simple network structure (Structure 1), what interconnects all access ISPs?

A single global transit ISP

54. In the hierarchy of ISPs, what are the approximately dozen very large ISPs called?

Tier-1 ISPs

55. What does it mean for an ISP to multi-home?

To connect to two or more provider ISPs

56. What is the primary benefit of multi-homing?

It allows an ISP to continue sending and receiving packets even if one of its providers has a failure

57. What does it mean for two ISPs to peer?

They can directly connect their networks so that traffic between them passes over the direct connection rather than through upstream intermediaries

58. What is an IXP?

An Internet Exchange Point (IXP) is a meeting point where multiple ISPs can peer together

59. What is a key characteristic of content-provider networks like Google's?

They are private networks that attempt to "bypass" the upper tiers of the Internet by peering directly with lower-tier ISPs

1.4 Delay, Loss, and Throughput

60. What are the four types of delays that contribute to total nodal delay?

Nodal processing delay, queuing delay, transmission delay, and propagation delay

61. Which delay is defined as the time required to examine the packet's header and determine where to direct it?

Processing delay

62. Which delay depends on the number of other packets queued for transmission on the same link?

Queuing delay

63. What is the formula for transmission delay? (L: packet length, R: link transmission rate)

$d_{\text{trans}} = L / R$

64. What is the formula for propagation delay? (d: distance, s: propagation speed)

$d_{\text{prop}} = d / s$

65. What is the formula for total nodal delay?

$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$

66. Which delay component can be the dominant term for a geostationary satellite link?

Propagation delay

67. Which delay component can vary significantly from packet to packet?

Queuing delay

68. What is the term for the ratio λ/R ?

Traffic intensity

69. What happens to queuing delay if the traffic intensity is greater than 1?

The queue will tend to increase without bound and the queuing delay will approach infinity

70. What is the "golden rule" of traffic engineering regarding traffic intensity?

Design your system so that the traffic intensity is no greater than 1

71. What happens to a packet that arrives at a router to find a full queue?

It is dropped (lost)

72. For an end-to-end path with N-1 routers, and neglecting queuing and processing delays, what is the end-to-end delay?

$d_{\text{end-end}} = N \times (L / R) + (\text{sum of all } d_{\text{prop}})$

73. What is the name of the program that can be used to measure end-to-end delay?

Traceroute

74. In addition to delay and packet loss, what is another critical performance measure in computer networks?

End-to-end throughput

75. In a two-link network with rates R_s and R_c , what is the throughput for a file transfer?

The minimum of $\{R_s, R_c\}$

76. What is the term for the link that constrains the end-to-end throughput?

The bottleneck link

77. In a network with N links of rates $R_1, R_2, \dots, R_N$, what is the throughput?

$\min\{R_1, R_2, \dots, R_N\}$

78. In today's Internet, what is typically the constraining factor for throughput for a single download?

The access network

79. If multiple data flows share a common link, how can that link become the bottleneck even if its rate is high?

If the aggregate traffic rate of the flows passing through it approaches or exceeds its capacity, the throughput for each flow will be reduced

1.5 Protocol Layers and Their Service Models

80. Why do network designers organize protocols into layers?

To provide structure to the design of network protocols

81. What is the service model of a layer?

The services that a layer offers to the layer above it

82. How does a layer provide its service?

By (1) performing actions within that layer and (2) using the services of the layer directly below it

83. Where are application-layer protocols like HTTP typically implemented?

In software in the end systems

84. What is the name for the protocols of the various layers taken together?

The protocol stack

85. What are the five layers of the Internet protocol stack?

Application, Transport, Network, Link, and Physical

86. What is the packet of information at the application layer called?

A message

87. What are the two main transport-layer protocols in the Internet?

TCP and UDP

88. What kind of service does TCP provide?

A connection-oriented service with guaranteed delivery, flow control, congestion control, and segmentation

89. What kind of service does UDP provide?

A connectionless, no-frills service that provides no reliability, no flow control, and no congestion control

90. What is the packet at the transport layer called?

A segment

91. What is the primary responsibility of the network layer?

To move network-layer packets (datagrams) from one host to another

92. What is the name of the network-layer packet in the Internet?

A datagram

93. What is the celebrated protocol at the Internet's network layer?

The Internet Protocol (IP)

94. What does the network layer contain in addition to the IP protocol?

Routing protocols

95. What service does the link layer provide to the network layer?

It moves a packet from one node (host or router) to the next node in the route

96. What is the name of the link-layer packet?

A frame

97. What is the job of the physical layer?

To move the individual bits within the frame from one node to the next

98. Which layers does a router typically implement?

Layers 1 through 3 (Physical, Link, and Network)

99. Which layers does a link-layer switch typically implement?

Layers 1 and 2 (Physical and Link)

100. What is the important concept illustrated as data moves down the protocol stack?

Encapsulation

101. At each layer, what are the two types of fields a packet has?

Header fields and a payload field

102. What does the payload of a layer typically contain?

The packet from the layer above

1.6 Networks Under Attack

103. What is the collective term for malicious software that can infect our devices via the Internet?

Malware

104. What is a network of thousands of compromised devices controlled by bad guys called?

A botnet

105. What does self-replicating malware do?

Once it infects one host, it seeks entry into other hosts over the Internet, spreading exponentially

106. What does a Denial-of-Service (DoS) attack do?

It renders a network, host, or other infrastructure unusable by legitimate users

107. What are the three categories most Internet DoS attacks fall into?

Vulnerability attack, Bandwidth flooding, and Connection flooding

108. What is a DDoS attack?

A Distributed Denial-of-Service attack, where the attacker controls multiple sources to blast traffic at a target

109. What is packet sniffing?

The act of recording a copy of every packet that flies by, often done on wireless networks

110. Why are packet sniffers difficult to detect?

Because they are passive and do not inject packets into the channel

111. What is one of the best defenses against packet sniffing?

Cryptography

112. What is it called when someone creates a packet with a false source address?

IP spoofing

113. What mechanism is needed to be sure a message originates from where we think it does?

End-point authentication

1.7 History of Computer Networking and the Internet

114. By the end of the 1970s, which three key Internet protocols were conceptually in place?

TCP, UDP, and IP

115. What was the flag day for the official deployment of TCP/IP on the ARPAnet?

January 1, 1983

116. What protocol did TCP/IP replace on the ARPAnet?

The NCP protocol

117. What does DNS stand for, and what is its purpose?

Domain Name System. It maps human-readable Internet names to IP addresses

118. What was the name of the ambitious French data networking project launched in the early 1980s?

The Minitel project

119. What are the four "killer applications" listed for the Internet by the end of the 1990s?

E-mail, the Web, Instant messaging, and Peer-to-peer file sharing

120. What has high-speed home Internet access set the stage for?

A wealth of video applications (YouTube, Netflix, video conferencing)

121. In what year did the number of wireless devices connecting to the Internet surpass the number of wired devices?

2011

122. What have online service providers like Google deployed to bypass the Internet as much as possible?

Their own extensive private networks, which peer directly with lower-tier ISPs

123. Where are many Internet commerce companies now running their applications?

In the "cloud" (e.g., Amazon EC2, Microsoft Azure, Alibaba Cloud)

1.8 Summary & Road-Mapping

124. According to the book's roadmap, which chapters are considered the five core chapters?

Chapters 2 through 6

125. Which layers of the Internet protocol stack do the five core chapters cover?

The top four layers: Application, Transport, Network, and Link

126. What topics are covered in Chapter 7?

Wireless and Mobile Networks (Wireless LANs, Cellular, Mobility)

127. What does Chapter 8 address?

Security in Computer Networks

Deep Dive & Conceptual Questions

128. What is the fundamental difference between how resources are managed in packet switching versus circuit switching?

In packet switching, resources are not reserved; they are used on demand, which can lead to congestion. In circuit switching, resources are reserved for the entire duration of the session

129. Why is packet switching considered more efficient for bursty data traffic?

Because it allows multiple data flows to share the same links and switches statistically, rather than dedicating a fixed share of resources to a single flow that may be idle at times

130. How does the concept of "statistical multiplexing" relate to packet switching?

Statistical multiplexing is the core principle of packet switching, where packets from different sources are interleaved over a shared link, without a guaranteed reservation of bandwidth for any single source

131. What is the relationship between a host, an end system, and a server?

A host is the same as an end system. A server is a type of host that typically provides a service (e.g., a web server). Not all hosts are servers; some are clients

132. Why is the layering principle criticized by some? (Name two potential drawbacks)

1. One layer may duplicate lower-layer functionality
2. Functionality at one layer may need information that is present only in another layer, violating the goal of separation

133. In the context of access networks, what does "asymmetric" mean?

The downstream transmission rate (from the ISP to the home) is typically much higher than the upstream rate (from the home to the ISP)

134. How does a router differ from a link-layer switch in terms of the layers they implement?

A router implements layers 1-3 (Physical, Link, Network) and can forward packets based on IP addresses. A link-layer switch implements only layers 1-2 (Physical, Link) and forwards frames based on MAC addresses

135. What is the key difference between transmission delay and propagation delay?

Transmission delay is the time it takes to push all of the packet's bits into the link. Propagation delay is the time it takes for a single bit to travel from the start of the link to the end of the link

136. If the traffic intensity is 0, what can be said about the average queuing delay?

It will be close to zero, as packet arrivals are infrequent and an arriving packet is unlikely to find another packet in the queue

137. Why does the end-to-end throughput for a file transfer equal the rate of the bottleneck link?

Because the source cannot send data faster than the slowest link in the path can carry it away. The bottleneck link constrains the entire flow, like the narrowest point in a pipe constrains water flow

138. How does encapsulation help in the layered architecture?

It allows each layer to add its own control information (header) without affecting the data from the layer above, enabling functions like addressing, error detection, and protocol-specific actions at each layer

139. Why is a DDoS attack more potent than a simple DoS attack from a single host?

Because it aggregates traffic from many controlled sources, making it easier to generate enough traffic to overwhelm a high-capacity target and making it harder to filter out the attack traffic

140. What is the significance of the transition from NCP to TCP/IP?

It marked the beginning of the modern Internet, establishing the robust and scalable TCP/IP protocol suite as the foundation for global internetworking

End of Study Guide

Wish you all the best!

Fathy.